

BARATH TIRUMURUHAN

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SUMMARY

4th-year Materials Science PhD specializing in optical and materials process modeling using Lumerical (FDTD & STACK) and MATLAB. Experienced in simulation, DOE, and optimization of thin-film and device structures. Versatile researcher with proven ability to adapt across disciplines, from optoelectronic devices to biological nanomaterials, to accelerate process development and cross-functional R&D innovation. **Available for summer 2026 internships and full-time positions from September 2027**

EDUCATION

- University of Minnesota (UMN)** Sep 2022 - Sep 2027 (Expected)
Ph.D. Materials Science & Engineering
◦ GPA: 3.68/4
Relevant Courses: Intro to Modern Optics, Electronic Properties of Materials, Mechanical Properties of Materials, Applied Machine Learning, Nonparametric Statistics
- Birla Institute of Technology & Science (BITS), Pilani** Aug 2017 - May 2022
M.Sc. Chemistry; B.E. Mechanical Engineering
◦ GPA: 9.2/10
Relevant Courses: Electrochemistry, Solid-state chemistry, Chemical & Biological Sensors

TECHNICAL SKILLS

- Modeling & Simulation:** Lumerical (FDTD & STACK), MATLAB, Python, Thin-film Optics, Process Modeling, Data Acquisition & Analysis
- Statistics & Data Analysis:** Design of Experiments (DOE), Nonparametric Statistics, Process Optimization
- Machine Learning:** Python, R, Scikit-Learn, Pytorch
- Materials Science & Devices:** OLEDs, Optoelectronics, Nanoparticle Formulations, Li-ion Batteries
- Analytical Characterization:** Spectroscopy (UV-Vis & Fluorescence, NMR), Chromatography (GC, HPLC), Electrochemistry (CV, Impedance Spectroscopy)
- Design & Prototyping:** Process Development, SolidWorks, Fusion 360

RESEARCH EXPERIENCE

- PhD Researcher - OLED Device Simulation (Ferry Group)** June 2025 - Present
UMN Minneapolis, Minnesota
◦ Model optical interference and emission profiles using Lumerical(FDTD & STACK) and MATLAB to guide OLED design
◦ Collaborate with experimental team to guide design improvements and validate prototypes
◦ Work under NDA on next-generation OLED product development
- PhD Researcher – Nanoparticle Barcoding (Boehnke Group)** Sep 2022 - Apr 2025
UMN Minneapolis, Minnesota
◦ Streamlined synthesis and bioanalytical workflows, improving reproducibility of new NP barcoding platform
◦ Developed and prototyped new in vitro cell assay protocols, integrating conjugation, formulation, extraction, and GC-MS analysis to evaluate nanoparticle–tumor interactions
- Undergraduate Research Intern (Dr. Naga Phani B Aetukuri)** Jun 2021 - May 2022
Indian Institute of Science Bengaluru, Karnataka
◦ Built a coulometry-based battery lifetime prediction framework, integrating Python and LabView for automated charge-discharge test data collection, modeling, and reliability analysis
◦ Designed and prototyped custom pressure-cell assemblies to evaluate stack pressure effects on garnet-type solid electrolytes with interlayers
- Undergraduate Research Assistant (Dr. Sujith R)** Jun 2019 - May 2021
BITS, Pilani Hyderabad, Telangana
◦ Optimized boron carbide reinforced Al-Zn-Mg-Cu alloy for ballistic armors using DOE & Taguchi analysis
◦ Engineered silicon and silicon oxycarbide-based anode materials for high-capacity Li-ion batteries

PUBLICATIONS

- Karmaker, S., Rosales, P. D., **Tirumuruhan, B.**, Viravalli, A., & Boehnke, N. (2025). **More than a delivery system: The evolving role of lipid-based nanoparticles.** In *Nanoscale*, RSC, Vol. 17(19), pp. 11864–11893. DOI: 10.1039/D4NR04508D (*All authors contributed equally; listed alphabetically. Last author is corresponding author*)
- Sharma, A., **Tirumuruhan, B.**, Muthuvel, G. S., Gupta, A. K., & Sujith, R. (2022). **Optimization of Process Parameters of Boron Carbide-Reinforced Al-Zn-Mg-Cu Matrix Composite Produced by Pressure-Assisted Sintering.** In *Journal of Materials Engineering and Performance*, Vol. 31, Issue 1, pp. 328-340. DOI: 10.1007/s11665-021-06210-4
- Gangadhar, J., **Tirumuruhan, B.**, & Sujith, R. (2022). **Applications and Future Trends in Mesoporous Materials.**, In A. Uthaman, S. Thomas, T. Li, & H. Maria (Eds.), *Advanced Functional Porous Materials: From Macro to Nano Scale Lengths*, Springer International Publishing, pp. 235-258. DOI: 10.1007/978-3-030-85397-6_8

HONORS AND AWARDS

- **Outstanding Teaching Assistant (Fall 2025)** May 2026
MATS 4221: Materials Performance, UMN Minneapolis, Minnesota
- **Outstanding Teaching Assistant (Fall 2023)** May 2024
MATS 2002: Introductory Materials Engineering Lab, UMN Minneapolis, Minnesota
- **Rutherford Aris, Usha Kumar, & George Philippidis Fellow** Sep 2022 - May 2023
Chemical Engineering & Materials Science (CEMS), UMN Minneapolis, Minnesota
- **University Merit List, BITS, Pilani)** May 2022
Recognized for academic performance among top 2% of students Hyderabad, Telangana
- **Certificate of Merit in Chemistry, CBSE** May 2017
Top 0.1% of national candidates in All India Senior Secondary Exam Chennai, Tamil Nadu

LEADERSHIP & SERVICE

- **Member of Graduate Student Committee (GSC)** Mar 2023 - Present
CEMS, UMN
 - Represented PhD cohort in department-wide decisions. Collaborated with faculty and peers to strengthen training and community engagement
- **Research Group Manager** May 2023 - Apr 2025
Boehnke Group
 - Directed procurement and inventory operations. Served as Deputy Lab Safety Officer, ensuring compliance with University Health & Safety standards
- **Moderator of RAMP Workshop** Feb 2025
Division of Chemical Health and Safety (DCHAS), ACS
 - Facilitated discussions on hazard recognition, risk assessment, and safe laboratory practices for PhD students across US